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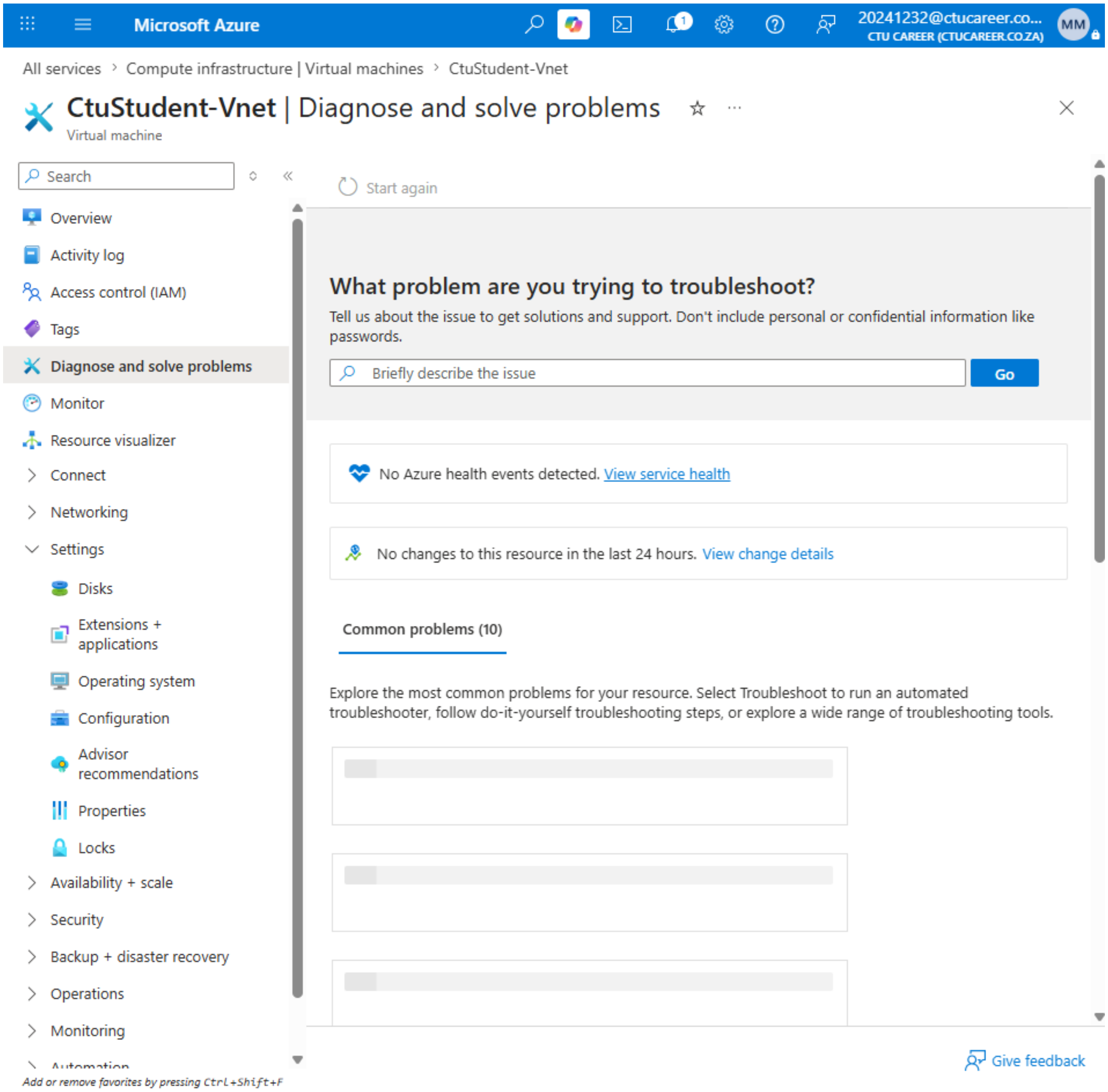
Practical Week 7

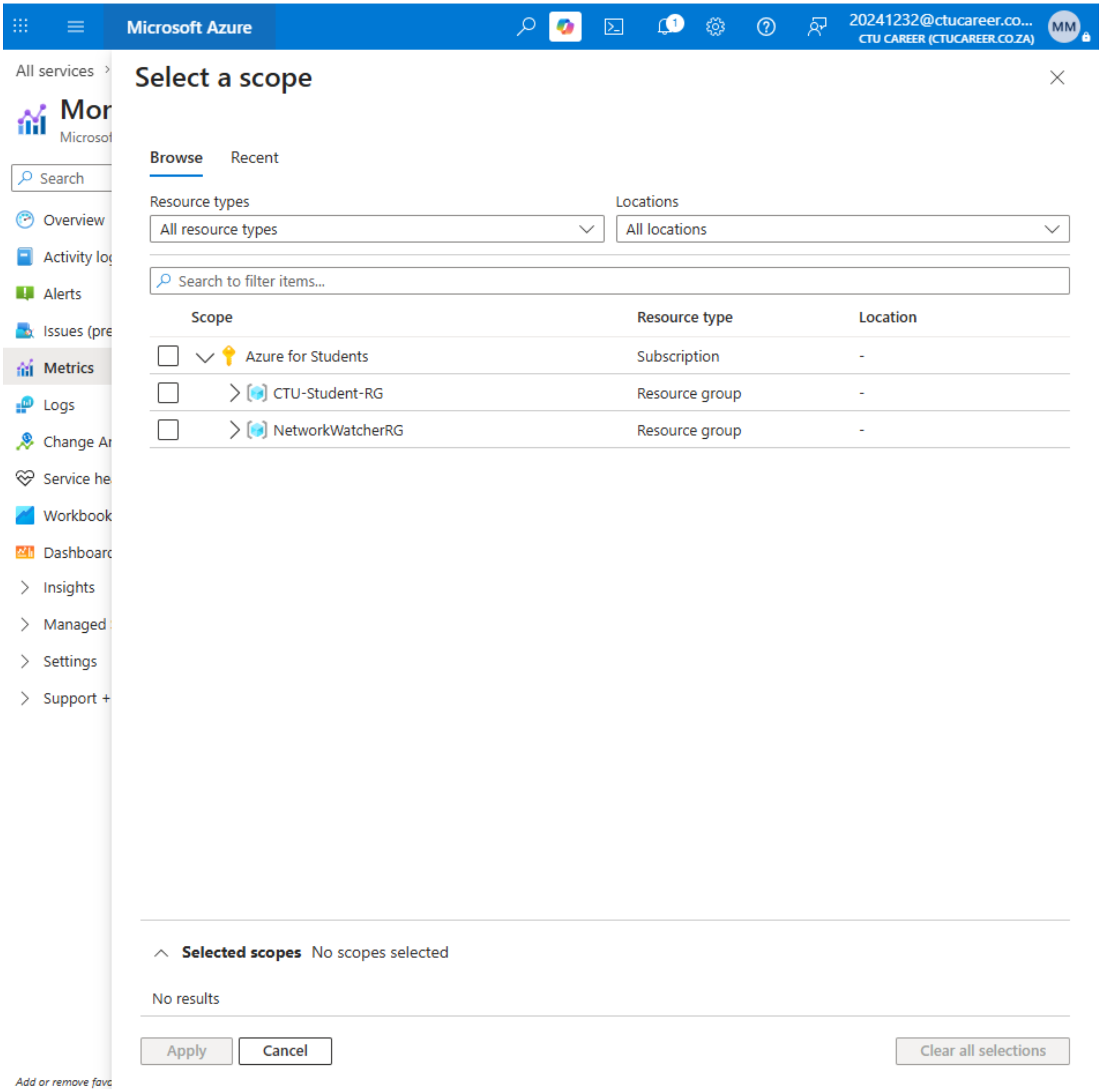
Task 1: Enable Azure Monitor



This is where I would have completed the practical if my subscription was active.

I would have enabled VM Insights and Diagnostic settings on my virtual machine to begin collecting essential telemetry like CPU, memory, and disk performance data. For my network resources, I would have configured diagnostic settings on the Virtual Network and Network Security Group to track traffic flow and security events, directing this data to a Log Analytics Workspace for centralized storage. I would have ensured that the correct log categories, such as network security group rule counters, were selected to provide a comprehensive view of the environment's health and security posture.

Task 2: Review Key Metrics



I would have used the Metrics Explorer to monitor CPU usage, selecting a specific time range to identify performance trends under different workloads. To track network throughput, I would have created a single chart showing both inbound and outbound traffic in bytes or bits per second to visualize the data flow patterns. Finally, I would have reviewed disk metrics by looking at read and write operations to understand how the system handles data-intensive tasks and to identify any potential storage bottlenecks.

Task 3: Document Your Monitoring Setup

I would have documented my configuration by listing every monitored resource and the specific data collection methods used, such as VM Insights for the server and diagnostic settings for the networking components. My summary would have explained that the data is stored in a Log Analytics Workspace, where it remains accessible based on the defined retention policy for historical analysis. I would have also reflected on the metrics observed, noting that consistent performance is expected and explaining how production anomalies, like unexpected traffic spikes, would be investigated using these tools.

Reflection and Alerts

I would have configured an alert rule to trigger a notification if CPU usage exceeded eighty percent for five minutes, allowing for proactive intervention before a performance issue impacted users. In my reflection, I would have explained that cloud monitoring is vital for visibility into virtualized infrastructure and that metrics are better for real-time performance tracking while logs are essential for deep-dive troubleshooting and security audits. I also would have noted that monitoring data is a key tool for cost optimization, as it reveals underutilized resources that could be resized to save money.